

CO4-ACTIVITY-CORRELATION AND OUTLIERS

ACTIVITY: Data Analysis Using Student Database

Aim

To perform data analysis on a student database by identifying correlations, detecting outliers, creating groupings, analyzing data, and filtering relevant records using Python in Google Colab.

Software Requirements

- Google Colab
 - Python 3
 - Pandas
 - NumPy
 - Matplotlib
 - SciPy
-

Dataset Used

Student Database (student_database.csv) containing the following attributes:

- Student ID
- Name
- Gender
- Department
- Year
- Study Hours
- Attendance Percentage
- Assignment Score
- Internal Marks
- Final Marks

- CGPA
- Projects Completed
- Placement Training

```
Student_ID,Name,Gender,Department,Year,Study_Hours,Attendance_Percentage,Assignment_Score,Internal_Marks,Final_Marks,CGPA,Projects_completed,Placement_Training
S101,Aarav,Male,CSE,2,2,72,58,60,62,6.8,1,No
S102,Diya,Female,CSE,2,4,81,70,72,74,7.5,2,Yes
S103,Rahul,Male,AI&DS,3,6,90,85,84,88,8.7,3,Yes
S104,Priya,Female,AI&DS,3,5,88,80,82,84,8.5,2,Yes
S105,Karthik,Male,ECE,2,3,76,65,67,69,7.1,1,No
S106,Sneha,Female,ECE,3,7,95,91,90,94,9.3,4,Yes
S107,Arjun,Male,CSE,4,8,96,94,95,97,9.7,5,Yes
S108,Meena,Female,EEE,2,2,70,55,57,59,6.5,1,No
S109,Vikram,Male,EEE,3,5,85,78,80,81,8.1,2,Yes
S110,Ananya,Female,IT,4,6,92,88,89,91,9.0,3,Yes
S111,Rohan,Male,IT,2,3,74,62,64,65,6.9,1,No
S112,Keerthi,Female,CSE,1,4,83,72,73,75,7.6,2,Yes
S113,Yash,Male,AI&DS,1,1,65,45,48,50,5.8,0,No
S114,Nisha,Female,ECE,4,7,94,92,91,95,9.5,4,Yes
S115,Harish,Male,CSE,3,5,86,79,81,83,8.3,2,Yes
S116,Divya,Female,IT,3,6,89,84,86,87,8.8,3,Yes
S117,Sanjay,Male,EEE,4,9,98,97,98,99,9.9,5,Yes
S118,Pooja,Female,AI&DS,2,4,82,73,75,76,7.7,2,Yes
S119,Manoj,Male,ECE,1,2,68,50,52,54,6.0,1,No
S120,Kavya,Female,CSE,4,10,99,100,99,100,10.0,6,Yes
S121,Aditya,Male,IT,2,3,77,66,67,68,7.0,1,No
S122,Lakshmi,Female,EEE,3,6,91,87,88,90,8.9,3,Yes
S123,Naveen,Male,AI&DS,4,8,97,95,96,98,9.8,5,Yes
S124,Aisha,Female,CSE,1,2,73,60,61,63,6.7,1,No
S125,Surya,Male,ECE,3,12,45,98,99,100,9.8,6,Yes
```

Task 1: Identifying Correlations

Objective

To determine the relationship between numerical variables in the student dataset.

Procedure

1. Import the required libraries.
2. Load the student dataset.
3. Select numerical columns.
4. Generate the correlation matrix.
5. Display the heatmap.
6. Plot a scatter graph between Study Hours and Final Marks.

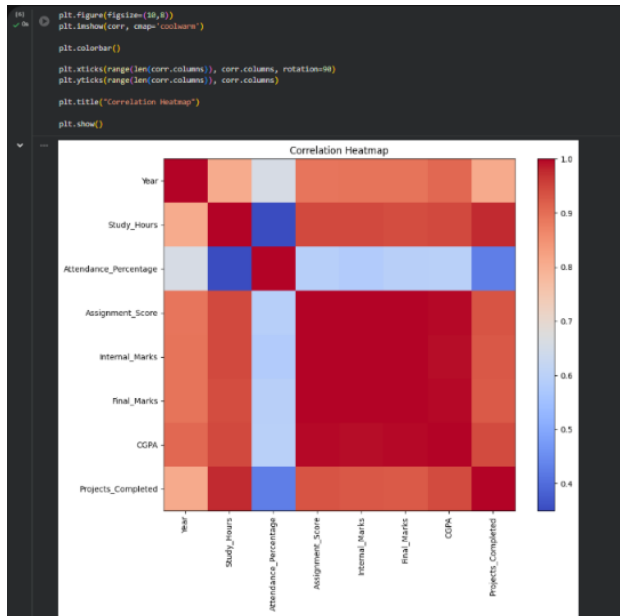
Output 1: Correlation Matrix

```
numeric_df = df.select_dtypes(include=np.number)
corr = numeric_df.corr()
corr
```

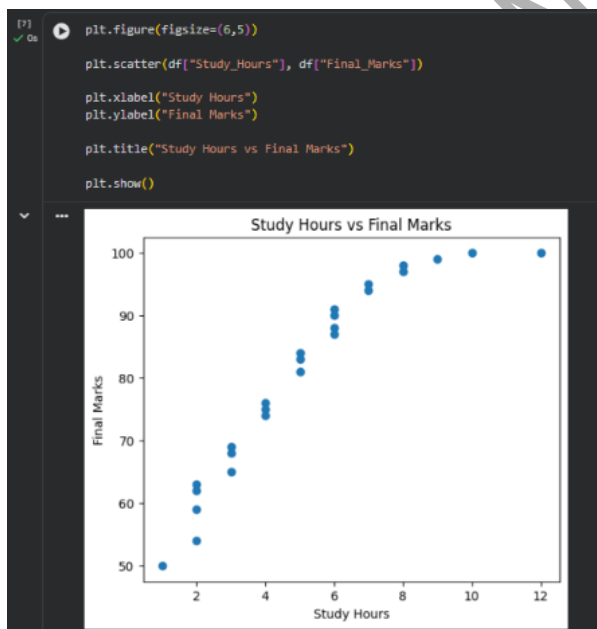
	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed
Year	1.000000	0.806129	0.659275	0.888544	0.891417	0.892953	0.907160	0.808473
Study_Hours	0.806129	1.000000	0.349495	0.944052	0.945551	0.938350	0.945604	0.976363
Attendance_Percentage	0.659275	0.349495	1.000000	0.591603	0.582646	0.595402	0.598422	0.427014
Assignment_Score	0.888544	0.944052	0.591603	1.000000	0.998447	0.998870	0.995275	0.932045
Internal_Marks	0.891417	0.945551	0.582646	0.998447	1.000000	0.998279	0.994721	0.928271
Final_Marks	0.892953	0.938350	0.595402	0.998870	0.998279	1.000000	0.996207	0.925695
CGPA	0.907160	0.945604	0.598422	0.995275	0.994721	0.996207	1.000000	0.941925
Projects_Completed	0.808473	0.976363	0.427014	0.932045	0.928271	0.925695	0.941925	1.000000

Next steps: [Generate code with corr](#) [New interactive sheet](#)

Output 2: Correlation Heatmap



Output 3: Scatter Plot (Study Hours vs Final Marks)



Observation

- Strong positive correlation exists between Study Hours and Final Marks.
- Attendance also shows a positive relationship with CGPA.

- Students who study more generally score higher marks.

Task 2: Identifying Outliers

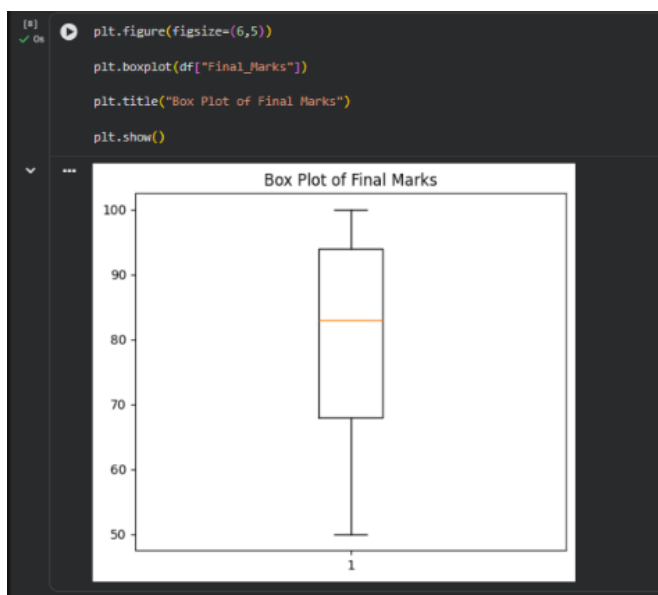
Objective

To identify unusual observations in the dataset.

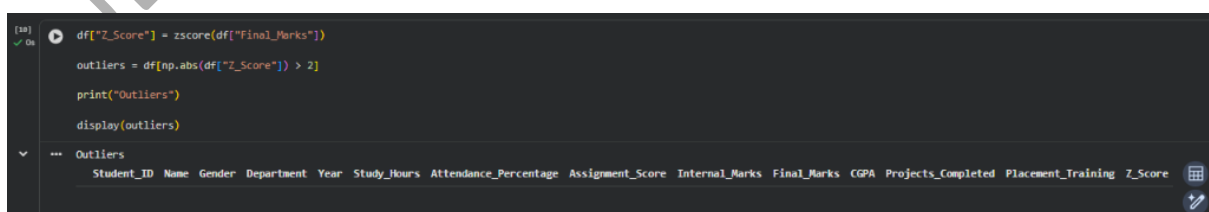
Procedure

1. Draw a box plot.
2. Calculate Z-score.
3. Identify records having Z-score greater than the threshold.

Output 1: Box Plot



Output 2: Z-score Outliers



Observation

- Most students fall within the normal range.

- A few students have exceptionally high or low marks.
- These observations are identified as outliers.

Task 3: Creating Groupings

Objective

To group students based on different categories.

Procedure

1. Group students by Department.
2. Calculate average Final Marks.
3. Create a Pivot Table.
4. Display a bar chart.

Output 1: GroupBy Result

```
(11)
✓ Os      group = df.groupby("Department")["Final_Marks"].mean()
          print(group)

v
Department
AI&DS    79.200000
CSE      79.142857
ECE      82.400000
EEE      82.250000
IT       77.750000
Name: Final_Marks, dtype: float64
```

Output 2: Bar Chart



Output 3: Pivot Table

```
[13] pivot = pd.pivot_table(
df,
values="Final_Marks",
index="Department",
columns="Gender",
aggfunc="mean"
)
pivot
```

	Gender	Female	Male
Department			
AIDS		80.0	78.666667
CSE		78.0	80.666667
ECE		94.5	74.333333
EEE		74.5	90.000000
IT		89.0	66.500000

Next steps: [Generate code with pivot](#) [New interactive](#)

Observation

- Departments have different average performances.
- Pivot tables simplify comparisons between male and female students.

Task 4: Analyzing Data

Objective

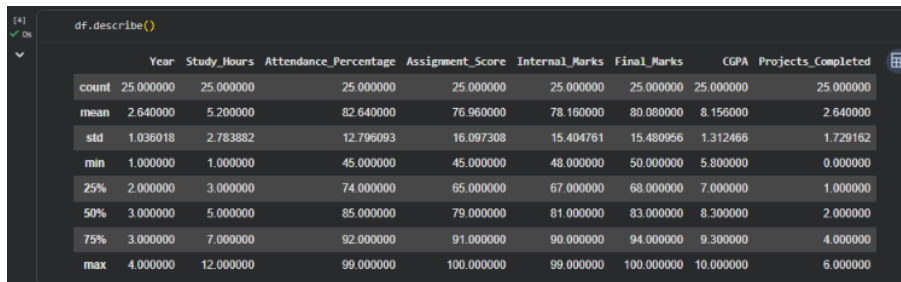
To summarize the student data using statistical methods and visualizations.

Procedure

1. Calculate Mean.
2. Calculate Median.

3. Calculate Mode.
4. Plot Histogram.
5. Plot Line Graph.
6. Plot Pie Chart.

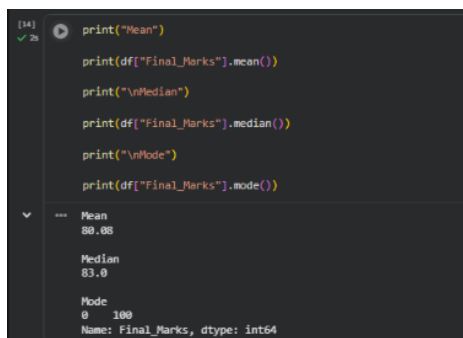
Output 1: Descriptive Statistics



A screenshot of a Jupyter Notebook cell showing the output of the `df.describe()` function. The output is a table with 10 rows and 9 columns. The columns are: count, Year, Study_Hours, Attendance_Percentage, Assignment_Score, Internal_Marks, Final_Marks, CGPA, and Projects_Completed. The rows represent various statistical measures: count, mean, std, min, 25%, 50%, 75%, and max.

	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed
count	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000	25.000000
mean	2.640000	5.200000	82.640000	76.960000	78.160000	80.080000	8.156000	2.640000
std	1.036018	2.783882	12.796093	16.097308	15.404761	15.480956	1.312466	1.729162
min	1.000000	1.000000	45.000000	45.000000	48.000000	50.000000	5.800000	0.000000
25%	2.000000	3.000000	74.000000	65.000000	67.000000	68.000000	7.000000	1.000000
50%	3.000000	5.000000	85.000000	79.000000	81.000000	83.000000	8.300000	2.000000
75%	3.000000	7.000000	92.000000	91.000000	90.000000	94.000000	9.300000	4.000000
max	4.000000	12.000000	99.000000	100.000000	99.000000	100.000000	10.000000	6.000000

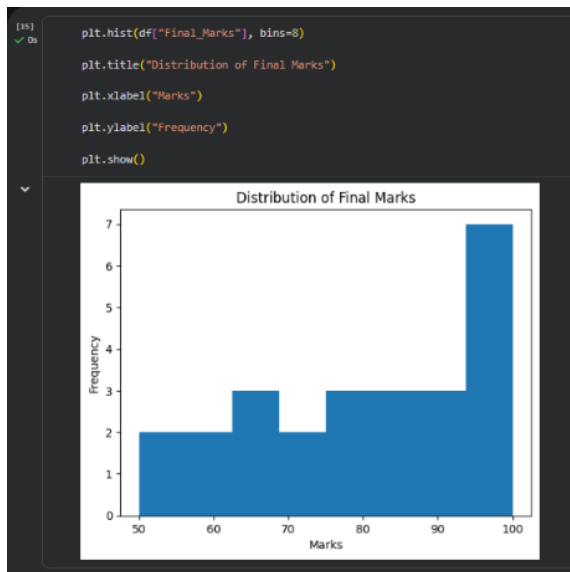
Output 2: Mean, Median and Mode



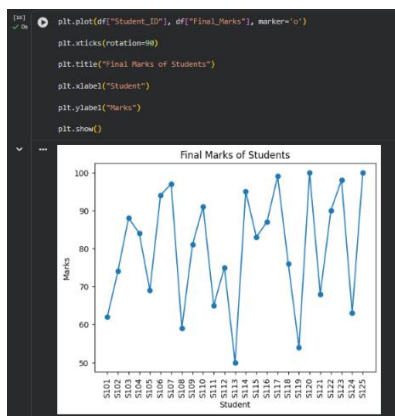
A screenshot of a Jupyter Notebook cell showing the calculation of mean, median, and mode for the 'Final_Marks' column. The code uses `print()` and `df['Final_Marks'].mean()`, `df['Final_Marks'].median()`, and `df['Final_Marks'].mode()`. The output shows the Mean as 80.08, the Median as 83.0, and the Mode as 0 and 100. The dtype for the mode is int64.

Statistic	Value
Mean	80.08
Median	83.0
Mode	0, 100

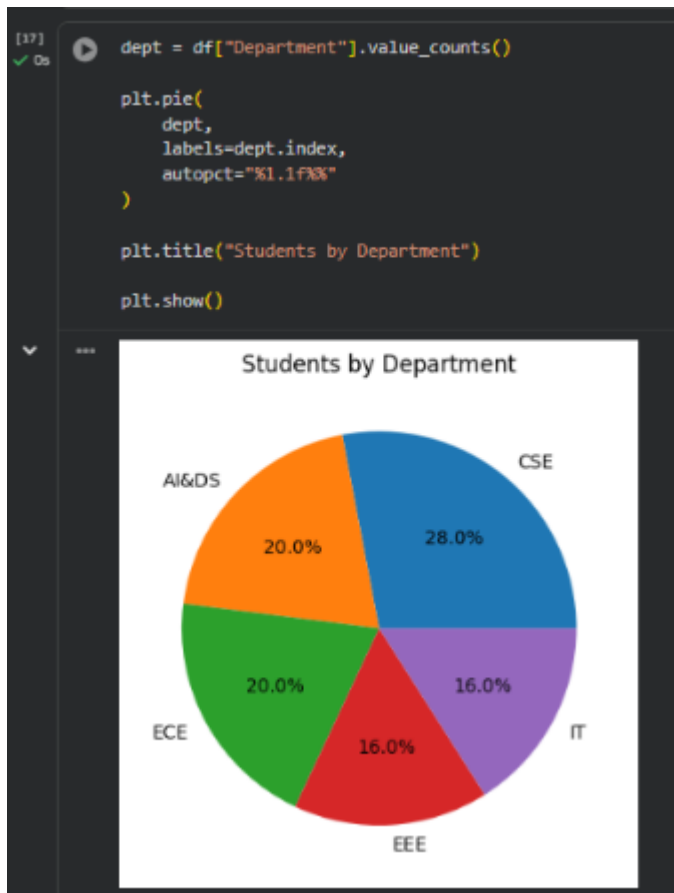
Output 3: Histogram



Output 4: Line Graph



Output 5: Pie Chart



Observation

- The average marks indicate overall student performance.
- Histogram shows the distribution of marks.
- Pie chart represents department-wise student distribution.
- Line graph shows variation in student marks.

Task 5: Separating and Focusing the Data

Objective

To filter the required student records for detailed analysis.

Procedure

1. Filter students with Final Marks greater than 90.
2. Filter students with Attendance above 90%.
3. Filter AI&DS students.

4. Filter students having CGPA greater than 8.5.
5. Sort students by Final Marks.

Output 1: Students with Final Marks > 90

```
[14]: top_students = df[df["Final_Marks"] > 90]
display(top_students)
```

	Student_ID	Name	Gender	Department	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed	Placement_Training	Z_Score
5	S106	Sneha	Female	ECE	3	7	95	91	90	94	9.3	4	Yes	0.917711
6	S107	Ajun	Male	CSE	4	8	96	94	95	97	9.7	5	Yes	1.115493
9	S110	Ananya	Female	IT	4	6	92	88	89	91	9.0	3	Yes	0.719928
13	S114	Nisha	Female	ECE	4	7	94	92	91	95	9.5	4	Yes	0.983638
16	S117	Sanjay	Male	EEE	4	9	98	97	98	99	9.9	5	Yes	1.247348
19	S120	Kavya	Female	CSE	4	10	99	100	99	100	10.0	6	Yes	1.313276
22	S123	Naveen	Male	AI&DS	4	8	97	95	96	98	9.8	5	Yes	1.181421
24	S125	Surya	Male	ECE	3	12	45	98	99	100	9.8	6	Yes	1.313276

Next steps: [Generate code with top_students](#) [New interactive sheet](#)

Output 2: Attendance > 90%

```
[15]: attendance = df[df["Attendance_Percentage"] > 90]
display(attendance)
```

	Student_ID	Name	Gender	Department	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed	Placement_Training	Z_Score
5	S106	Sneha	Female	ECE	3	7	95	91	90	94	9.3	4	Yes	0.917711
6	S107	Ajun	Male	CSE	4	8	96	94	95	97	9.7	5	Yes	1.115493
9	S110	Ananya	Female	IT	4	6	92	88	89	91	9.0	3	Yes	0.719928
13	S114	Nisha	Female	ECE	4	7	94	92	91	95	9.5	4	Yes	0.983638
16	S117	Sanjay	Male	EEE	4	9	98	97	98	99	9.9	5	Yes	1.247348
19	S120	Kavya	Female	CSE	4	10	99	100	99	100	10.0	6	Yes	1.313276
21	S122	Lakshmi	Female	EEE	3	6	91	87	88	90	8.9	3	Yes	0.654001
22	S123	Naveen	Male	AI&DS	4	8	97	95	96	98	9.8	5	Yes	1.181421

Next steps: [Generate code with attendance](#) [New interactive sheet](#)

Output 3: AI&DS Students

```
[16]: aids = df[df["Department"] == "AI&DS"]
display(aids)
```

	Student_ID	Name	Gender	Department	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed	Placement_Training	Z_Score
2	S103	Rahul	Male	AI&DS	3	6	90	85	84	88	8.7	3	Yes	0.522146
3	S104	Priya	Female	AI&DS	3	5	88	80	82	84	8.5	2	Yes	0.258436
12	S113	Yash	Male	AI&DS	1	1	65	45	48	50	5.8	0	No	-1.983099
17	S118	Pooja	Female	AI&DS	2	4	82	73	75	76	7.7	2	Yes	-0.268984
22	S123	Naveen	Male	AI&DS	4	8	97	95	96	98	9.8	5	Yes	1.181421

Next steps: [Generate code with aids](#) [New interactive sheet](#)

Output 4: CGPA > 8.5

```
[22] In: cpga = df[df["CGPA"] > 8.5]
      display(cpga)
```

	Student_ID	Name	Gender	Department	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed	Placement_Training	Z_Score
2	S103	Rahul	Male	AI&DS	3	6	90	85	84	88	8.7	3	Yes	0.522146
5	S106	Sneha	Female	ECE	3	7	95	91	90	94	9.3	4	Yes	0.917711
6	S107	Arjun	Male	CSE	4	8	96	94	95	97	9.7	5	Yes	1.115493
9	S110	Ananya	Female	IT	4	6	92	88	89	91	9.0	3	Yes	0.719928
13	S114	Nisha	Female	ECE	4	7	94	92	91	95	9.5	4	Yes	0.983638
15	S116	Divya	Female	IT	3	6	89	84	86	87	8.8	3	Yes	0.456218
16	S117	Sanjay	Male	EEE	4	9	98	97	98	99	9.9	5	Yes	1.247348
19	S120	Kavya	Female	CSE	4	10	99	100	99	100	10.0	6	Yes	1.313276
21	S122	Lakshmi	Female	EEE	3	6	91	87	88	90	8.9	3	Yes	0.654001
22	S123	Naveen	Male	AI&DS	4	8	97	95	96	98	9.8	5	Yes	1.181421
24	S125	Surya	Male	ECE	3	12	45	98	99	100	9.8	6	Yes	1.313276

Next steps: [Generate code with cpga](#) [New interactive sheet](#)

Output 5: Sorted Student Records

```
[23] In: sorted_df = df.sort_values(
      by="Final_Marks",
      ascending=False
    )
      display(sorted_df)
```

	Student_ID	Name	Gender	Department	Year	Study_Hours	Attendance_Percentage	Assignment_Score	Internal_Marks	Final_Marks	CGPA	Projects_Completed	Placement_Training	Z_Score
24	S125	Surya	Male	ECE	3	12	45	98	99	100	9.8	6	Yes	1.313276
19	S120	Kavya	Female	CSE	4	10	99	100	99	100	10.0	6	Yes	1.313276
16	S117	Sanjay	Male	EEE	4	9	98	97	98	99	9.9	5	Yes	1.247348
22	S123	Naveen	Male	AI&DS	4	8	97	95	96	98	9.8	5	Yes	1.181421
6	S107	Arjun	Male	CSE	4	8	96	94	95	97	9.7	5	Yes	1.115493
13	S114	Nisha	Female	ECE	4	7	94	92	91	95	9.5	4	Yes	0.983638
5	S106	Sneha	Female	ECE	3	7	95	91	90	94	9.3	4	Yes	0.917711
9	S110	Ananya	Female	IT	4	6	92	88	89	91	9.0	3	Yes	0.719928
21	S122	Lakshmi	Female	EEE	3	6	91	87	88	90	8.9	3	Yes	0.654001
2	S103	Rahul	Male	AI&DS	3	6	90	85	84	88	8.7	3	Yes	0.522146
15	S116	Divya	Female	IT	3	6	89	84	86	87	8.8	3	Yes	0.456218
3	S104	Priya	Female	AI&DS	3	5	88	80	82	84	8.5	2	Yes	0.258436
14	S115	Harish	Male	CSE	3	5	86	79	81	83	8.3	2	Yes	0.192508
8	S109	Vikram	Male	EEE	3	5	85	78	80	81	8.1	2	Yes	0.060653
17	S118	Pooja	Female	AI&DS	2	4	82	73	75	76	7.7	2	Yes	-0.268984
11	S112	Keerthi	Female	CSE	1	4	83	72	73	75	7.6	2	Yes	-0.334912
1	S102	Diya	Female	CSE	2	4	81	70	72	74	7.5	2	Yes	-0.400839
4	S105	Karthik	Male	ECE	2	3	76	65	67	69	7.1	1	No	-0.730477
20	S121	Aditya	Male	IT	2	3	77	66	67	68	7.0	1	No	-0.796404
10	S111	Rohan	Male	IT	2	3	74	62	64	65	6.9	1	No	-0.994187
23	S124	Aisha	Female	CSE	1	2	73	60	61	63	6.7	1	No	-1.126042
0	S101	Aarav	Male	CSE	2	2	72	58	60	62	6.8	1	No	-1.191969
7	S108	Meena	Female	EEE	2	2	70	55	57	59	6.5	1	No	-1.389752
18	S119	Manoj	Male	ECE	1	2	68	50	52	54	6.0	1	No	-1.719389
12	S113	Yash	Male	AI&DS	1	1	65	45	48	50	5.8	0	No	-1.983099

Observation

- Filtering helps identify high-performing students.
- Sorting arranges records in descending order of marks.
- Focused datasets improve analysis and decision-making.

Result

The student dataset was successfully analyzed using Python in Google Colab. Correlations between variables were identified, outliers were detected using box plots and Z-scores, students were grouped by department, descriptive statistical analysis and visualizations were performed, and relevant records were filtered and sorted. The experiment demonstrated how data analysis techniques can extract meaningful insights from educational data.

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